



Linnæus University

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Master's Thesis

In-Vehicle Information Systems

A Study on the Interaction between Driver and Automotive Head-Up Display System



Author: Shishengxiong Zhong

Supervisor: Niki Chatzipanagiotou

Examiner: Anita Mirijamdotter /Päivi Jokela

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Abstract and Keywords

Head-up display (HUD) systems are systems that present data on a transparent display without requiring the user of the system to look away from their viewpoints. Although, HUD systems are not a new technology, as they have been initially developed in 1950s for military aircrafts, the last decade are also used in the automotive industry. Automotive HUD compliments the driver's visual information as it projects important driving information onto the driver's windshield. The aim of the automotive HUD system is to provide information and assure the safety of the vehicle, however research studies show that it is sometimes considered a distraction for the drivers. Therefore, the purpose of this master's thesis research is to explore and improve the understanding of the interaction between the car driver and the automotive Head-Up Display system. Further, the master's thesis explores the suggestions of car drivers in order to improve their interaction with the automotive HUD system. For this, an interpretive qualitative focused-ethnographic research was conducted. The data was collected through passive observation and individual semi-structured face-to-face interviews with six purposively selected research participants-drivers of cars equipped with HUD system. The collected data was analysed thematically to conclude to six themes, which represent the research findings. These research findings were discussed with the help of the theoretical framework, which included main concepts from the human-computer interaction body of research, including User-Centred Design.

The research findings show that the participants receive visual information via the graphical interface of the HUD system and, at the same time, use the button interface on the steering wheel, and the touch screen on the centre console for input operations. The participants although overall satisfied by the interaction with the graphical interface, they experience some challenges. The findings show that the participants experience negative effects when disturbed by bright light, or misunderstood the information presented by the HUD system. However, most participants, especially those that are novice drivers, experience difficulties during their interactions regarding the input-based. Thus, the research contributes to the existing body of knowledge of the informatics research field and, in particular, the human-computer interaction body of research by providing insights regarding in-vehicle information systems and their interaction with the driver. The research also contributes to designers, developers, technology companies, and other interested stakeholders by offering insights and suggestions regarding the in-vehicle information systems based on a more user-centered design approach.

Keywords: In-Vehicle Information Systems, Automotive Head-Up Display System, Human-Computer Interaction, Augmented Reality, Interaction, Information, Safety, Focused-Ethnographic Research.

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Table of Contents

Abstract and Keywords	1
Acknowledgments	2
Table of Contents	3
List of Figures	4
List of Tables	4
Abbreviations	5
1. Introduction	6
1.1 Background and problem	6
1.2 Purpose Statement and Research Questions	7
1.3 Previous Research	8
1.4 Topic Justification	9
1.5 Scope and Limitations	9
1.6 Thesis Organization	10
2. Literature Review	12
2.1 Search strategy	12
2.2 Human-Computer Interaction	12
2.3 In-Vehicle Information Systems	13
2.4 Automotive Head-Up Display Systems	14
2.4.1 Technical Base: Augmented Reality	16
2.4.2 Interaction	16
2.4.3 Information	17
2.4.4 Safety	18
2.5 Theoretical framework: User-Centered Design	19
3. Methodology	20
3.1 Philosophical Tradition (Paradigm)	20
3.2 Research Approach and Methodology	21
3.3 Methods of Data collection	21
3.3.1 Sampling Technique, Criteria and Size	22
3.3.2 The method of Observation	22
3.3.3 The method of Interview	22
3.3.4 Observation and Interview Procedure	23
3.4 Method of Data Analysis	24
3.6 Validity and Reliability	25
3.7 Ethical Considerations	26
4. Empirical Findings	27
4.1 Overview of Findings	27

4.2 Theme 1: Forms of interaction	27
4.3 Theme 2: Setup and adaptation	28
4.4 Theme 3: Usage of dashboard and centre control display	29
4.5 Theme 4: Gaze and driving habits	29
4.6 Theme 5: Graphical interface	30
4.7 Theme 6: Suggestions to improve the interaction	31
4.8 Summary of Findings	31
5. Discussion	32
5.1 Answer to Research Question 1	32
5.2 Answer to Research Question 2	34
5.3 Discussion within the User-Centred Design theory	35
5.4 Suggestions of users regarding the interaction	36
6. Conclusion	38
6.1 Conclusions	38
6.2 Contributions	39
6.3 Suggestions for Future Research	39
References	41
Appendices	47
Appendix A: Informed Consent Form for Master's Thesis	47
Appendix B: Script and Questions for Semi-Structured Interviews	48

List of Figures

Figure 1. Organised Structure of Chapters	10
Figure 2. Automotive HUD system in action	14
Figure 3. Automotive HUD system vs Traditional HDD	15
Figure 4. Projection unit of automotive HUD system and its display	15
Figure 5. Projection unit of automotive HUD system and its display	19

List of Tables

Table 1. The arrangement with research participants	23
Table 2. Themes answering the 1st Research Question	32
Table 3. Theme answering the 2nd Research Question	34

Abbreviations

ACC: Autonomous Cruise Control

AR: Augmented Reality

AR-HUD: Augmented Reality Head-Up Display system

GPS: Global Positioning System

GUI: Graphical User Interface

HCI: Human–Computer Interaction

HDD: Head-Down Display

HUD: Head-Up Display

ICT: Information and Communications Technology

IS: Information System

ITS: Intelligent Transportation System

IoV: Internet of Vehicles

UCD: User-Centred Design

UX: User Experience

1. Introduction

In Chapter 1, an introduction to this master's thesis is presented. Begins by identifying the context and origin of the research question. Previous related studies in this field are presented to scope the research area and discuss the existing gaps. The importance of the study, the purpose of the research, and the research questions are stated. The limitations of the study are discussed. The chapter concludes by presenting the structure of this master's thesis accompanied by a figure.

1.1 Background and problem

Head-up display (HUD) systems are systems that present data on a transparent display without requiring the user of the system to look away from their viewpoints (Park, et al., 2013; Wang et al., 2018; Betancur et al., 2018). HUD systems are nothing new. They have been initially developed for military aircraft since the 1950s and are still in use (Park and Park, 2019). Thus, the origin of their name, which comes from the pilot of the aircrafts being able to view critical information, displayed on a transparent display that is positioned in front of them in the cockpit of the aircraft, with the head positioned up and forward, without looking at other instruments of the aircraft (Park et al., 2013; Wang et al., 2018; Belyavskii et al. 2021). Nowadays, HUDs are also used in commercial aircrafts and cars.

The head-up display systems used in cars, namely the automotive HUD, is a technology that allows the projection of an image onto the vehicle's windshield or a panel, just beneath the driver's line of sight (Bergman, 2011; Park et al., 2013; Ishizaki et al., 2014; Smith et al., 2015; Wang et al., 2018). It is provided to some cars, such as modern cars as an additional information and safety feature of the vehicle (Harvey et al., 2011; Park et al., 2013; Pauzie, 2015). That is, the automotive HUD provides and displays various information, such as hazard warnings on the road ahead, traffic sign notifications, or other driving information in relation to driving, without requiring the driver of the car to take their eyes off the road.

A lot of studies have shown that the most sensitive way for humans to receive information is first hearing, followed by touch, and finally vision (Park et al., 2013; Jain et al., 2015; Abbasi-Kesbi, Memarzadeh-Tehran, and Deen, 2017). Still, 80% of driving information comes from vision (Venkatasawmy, 2012). The sound may be ignored when the driver is fully absorbed in driving. The extra driving tactile sensation may interfere with the regular operation of the driver, except in some exceptional cases, such as vibration to wake up a tired driver (Park et al., 2013). Visual information is the primary source of information channel in driving (Abbasi-Kesbi, Memarzadeh-Tehran, and Deen, 2017). Hence, by displaying critical information to the driver and allowing them to maintain the maximum amount of time looking at the road, the automotive HUD system reduces the chance of accident through distraction, which makes it an additional safety feature of the vehicle. The addition of automotive HUDs has complimented the driver's visual information as it projects important driving information, such as information regarding an obstacle on the road ahead, onto the windshield, visible only from the driver's seat. Compared with reliance on the traditional head-down instrument cluster, automotive HUDs are regarded as an additional information feature of the vehicle because the provided and displayed information increases the time the driver spends to keep

the vision on the road, reduces the driver's energy or time spent on switching vision focus, as well as simplifies the combined tasks of driving safely (Park et al., 2013). In addition, it simplifies monitoring the vehicle speed and performance, navigating to the destination, even controlling the media players (Maurer et al., 2014).

Although the aim of the automotive HUD system is to provide information and assure the safety of the vehicle, research studies show that HUDs are sometimes considered a distraction for the drivers (Hilpert et al., 2014). A study by the University of Toronto found that the automotive HUD system may reduce driving safety (Wanneret al., 2016). The research conducted an experiment where participants were stimulated by multiple visual information. The experiment showed that the recognition and counting accuracy of stimulus points became significantly lower (Niakan et al., 2015). That is, drivers had to concentrate on what was happening on the road, around them as they always do, but they also had to attend to whatever warning popped up on the windshield in front of them. Another experimental study by the University of Central Florida concluded to similar research findings (Niakan et al., 2015). The research findings show that the speed at which the participants received information and gave reactions was significantly decreased (Lewis and Neider, 2016). Delays in receiving and responding to information may threaten people's driving. According to the Swedish traffic theory test, when the speed is 60 kilometers per hour, the car will move forward about 16-17 meters per second. A one-second delay could mean life or death while driving (Gao et al., 2021). The previous studies show that although automotive HUD systems were added to vehicles as an additional information and safety feature, they may have opposite results. Therefore, further research is needed to examine whether automotive HUD systems fulfil their initial purpose, information, and safety, or whether they are distracting and, consequently, putting in danger people's safety.

HUDs in-vehicle provide humans with a convenient and advanced service, but may also pose a considerable threat. Therefore, it's indispensable to study this technology in depth from the perspective of the user/driver. That is, investigate in detail the information HUD display to users while their driving, and identify the impact of its display method, presentation location, and information content on driving. Following the user's/driver's perspectives, a more user-centered design of HUD car systems is suggested.

1.2 Purpose Statement and Research Questions

This master's thesis, thus, focuses on the experiences and perspectives of car drivers driving cars equipped with automotive HUD systems. Thus, the purpose of the master's thesis is to explore and improve the understanding of the interaction between the car driver and the automotive Head-Up Display system. Further, the master's thesis explores the suggestions of car drivers in order to improve their interaction with the automotive HUD system.

This purpose statement has given rise to the following research questions:

RQ1: How do car drivers experience the use of the automotive HUD system in regard to their interaction with the respective system?

Sub-questions:

RQ1.1. *How do drivers of cars equipped with automotive HUD systems describe the benefits of using a HUD system?*

RQ1.2. *How do drivers of cars equipped with automotive HUD systems describe the challenges of using a HUD system?*

RQ2: What are the improvements that car drivers of cars equipped with HUD systems suggest for their interaction with the respective system?

1.3 Previous Research

In order to ensure that this research meets the academic context and satisfies the current academic demands, a systematic literature review is needed. After an initial reviewing of previous related studies that examine automotive HUD systems, very few studies were found to examine how car drivers experience the use of automotive HUD systems and, to my knowledge, no study was found to examine the improvements regarding the interaction with such a system from the car drivers' perspective. User-centric literature that closely observes users using HUDs, while actually driving, and interviews with drivers about their user experience seem were not found in the literature search. However, some related previous research studies were found and are briefly presented below.

Guo et al. (2014) conducted research to analyze drivers' attitudes towards the HUD system using a stated preference survey. In this study, the scholars investigated and ranked the users' preferences for needed information presented by the HUD, rendered style of HUDs' display, and projecting place. Although the research result shows the drivers' willingness about what kinds of functions the HUD systems should have and how to display the information, this study is based on a survey with some participants having a blank picture of the HUD system in vehicles. It does not involve any observations or investigations into the process of using the actual automotive HUD system while driving the vehicle in order to catch the participants' experiences.

Beck et al. (2018) conducted a study on the perceived importance of automotive HUD information items. During their study, they conducted a survey to participants who have more than 2 years' experience of using HUD systems in driving. The study aimed at listing a rank of the importance of information items that were displayed by existing commercial HUD systems. Since the participants have relevant experience, this study is a reliable reference. But it still has no practical observation or interview.

Beck, et al. (2019), conducted a study on the user experience of automotive HUD systems. Continuing their previous study, on the basis of clarifying important information items, they surveyed the feedback of experienced participants and gave optimization suggestions for each information item based on users' experiences. The study outcome shows how important information items on the HUD are optimized. Still, the outcome was not related to the actual driving process of cars equipped with HUD systems.

1.4 Topic Justification

With the improvement of technology, the automotive HUD system becomes more popular and advanced (Gabbard et al., 2014). Since General Motors released the first automotive HUD system in 1988 (Park and Park, 2019), commercial automotive HUDs have been used in a variety of vehicles (Pala, 2012; Betancur et al., 2018; Park and Park, 2019). The use rate of automotive HUDs is forecasted to increase in the coming years (Park et al., 2019). By 2024, nearly one-third of cars are expected to be equipped with HUD systems (Park et al., 2019). Automotive HUD capabilities are expanding as new technologies are invented in the areas such as optics, augmented reality, IoT, autonomous systems, etc. (Gabbard et al., 2014). Many previous studies have proposed various automotive HUD features that reflect its performance and technological advancements. For instance, the displaying hazard warnings (Charissis and Papanastasiou, 2008; George et al., 2012; Park and Kim, 2013; Lee et al., 2015), traffic sign notifications (Park et al., 2013; Lee et al., 2015; Yang et al., 2016), driving instructions (Park and Kim, 2013; Yoon et al., 2014; Lee et al., 2015). Although the technology to build the various functions of a HUD is feasible, it does not mean that automotive HUD systems can be designed to display an unlimited variety of information (Park and Park, 2019). Displaying too-much information through the HUD can lead to an information overload, which may cause cognitive impairment or distraction to the drivers (Pauzie, 2015; Sun, Wu, and Spence, 2015; Lewis and Neider, 2016; Beck and Park, 2018; Park and Park, 2019). Therefore, research from the user's perspective to explore and understand their experience and expectation regarding interaction of the automotive HUD system could be valuable. In addition, studying from this perspective could promote the development of HUD manufacturing in the automotive industry and brings more excellent user experience to the drivers.

The HUD systems that lead ahead nowadays are constructed based on Augmented Reality (Lewis and Neider, 2016). As is an emerging technology, the development of AR still needs various studies from different perspectives (Park et al., 2013). The research question proposed in this master's thesis is precisely aimed at an interesting area of AR applications - the automotive HUD system. However, new technologies always bring new experiences and challenges to people. IS researchers have been interested in the human-computer interaction of emerging technologies over the years. The focus of this master's thesis is essentially a human-computer interaction between the driver and the HUD system. The study on the HCI of such a new technology application could be beneficial to the informatics/information Systems research field. As aforesaid, this master's thesis could hence be of theoretical and practical contribution to conduct this research on in-vehicle HUD.

1.5 Scope and Limitations

The master's thesis explores and improves the understanding of the interaction between the car driver and the automotive Head-Up Display system. Further, the master's thesis explores the suggestions of car drivers in order to improve their interaction with the automotive HUD system.

The research only considers the user's -in this case the car drivers- experiences and perspectives, in regard to the use of the automotive HUD system. The manufacturers' perspective is out of the scope of the master's thesis research and, thus, they will not be reached and included in the study. Consequently, the automotive HUD systems will not be examined in relation to its technology (software and hardware)

The research will include only car drivers of cars equipped with an automotive HUD system in order to assure that the research participants have first-hand experience and, therefore, they will provide rich data to the researcher. In addition, part of the research will be conducted while the aforementioned car drivers drive their vehicles, aiming to catch the research participants' full experience and, thus, collect abundant and fruitful data.

Concluding, since the perspective of this research is inclined towards the public, it should not involve any moral or ethical issues for various stakeholders.

1.6 Thesis Organization

Figure 1 below represents the structure of this master's thesis. Although conducting research is not a linear process, for the sake of presentation and reporting, it is a linear flow diagram of different chapters organised from left to right in chronological and logical order, with the accompanying arrows depicting the relationship between the chapters. The background and research questions presented in the Introduction influenced the research's direction and the scope of the Literature Review. Then, Methodology is clarified to guide the data collection and analysis to produce findings. The findings are discussed with the Literature Review in the Discussion section. Based on both Introduction and Discussion, Conclusions are made.

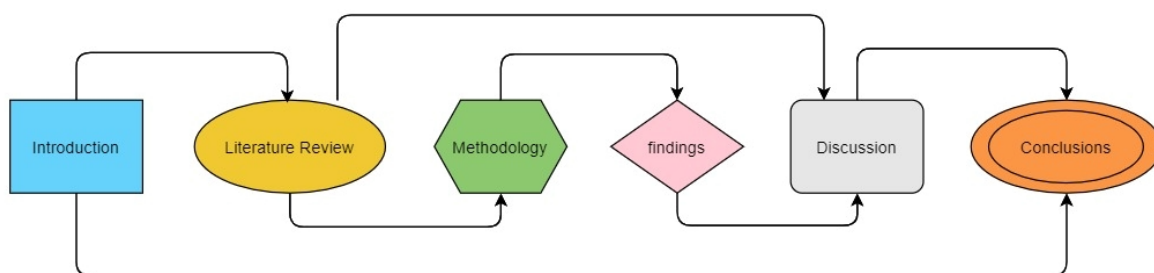


Figure 1. Organised Structure of Chapters

Chapter 1 presented an introduction to the master's thesis. It begun by identifying the context and origin of the research question. Previous related studies in this field were presented to scope the research area and discuss the existing gaps. The importance of the study, the purpose of the research and the research questions were stated. The limitations of the study were discussed. The chapter concluded by presenting the structure of this master's thesis accompanied by a figure.

Chapter 2 presents a systematic literature review. After clarifying the search strategy, the Human-Computer Interaction is presented as the body of research. Various in-vehicle information systems are discussed. The automotive HUD systems is focused as the key target of research, with a detailed literature review of its applications, technical basis, information provided, interaction methods, and safety. Lastly, the theoretical framework of this master's thesis is motivated.

Chapter 3 presents the research design of this master's thesis, which includes the selected paradigm and methodology. Subsequently, the methods for data collection and analysis are defined. The validity and reliability, as well as the ethical considerations are presented.

In Chapter 4, the empirical findings are presented. Firstly, an introduction of the themes is presented, together with their analysis and reflections. Each theme is presented in detail together with evidence, that is, participants' quotations and observation notes. The chapter concludes with a short summary of the findings.

In Chapter 5, a discussion based on the findings is conducted. The themes are categorized to answer each research question. All findings are discussed with the help of the theoretical framework. The suggestions and recommendations from the participants are listed.

As the conclusion, Chapter 6 summarises the overall outline of the master's thesis. The contributions of the master's thesis were presented. The related future work is motivated.

2. Literature Review

In chapter 2, a systematic literature review is presented. After clarifying the search strategy, the Human-Computer Interaction is presented as the body of research. Various in-vehicle information systems are discussed. The automotive HUD systems is focused as the key target of research, with a detailed literature review of its applications, technical basis, information provided, interaction methods, and safety. Lastly, the theoretical framework of this master's thesis is motivated.

2.1 Search strategy

A review of previous relevant studies is necessary to clarify and define the framework and concepts of this master's thesis research. To enhance academic rigor and value, a systematic review has been selected as the type of this literature review (Wanner et al., 2016), where search for specific terms and keywords in the designated databases of scholarly journals, magazines, and online libraries to initially explore the relevant literature, was conducted. A further selection was then performed based on filters such as publication year, language, degree of relevance, etc. to decide which is included and excluded.

In this literature review, the designated search databases were IEEE Xplore, complemented by Google Scholar. One Search was the used search engine from Linnaeus university. The scientific journals where the search was performed were: European Journal of Information Systems, Information Systems Journal, Information Systems Research, Journal of AIS, Journal of Information Technology, Journal of MIS, Journal of Strategic Information Systems, and MIS Quarterly. These journals are known as the top journals in the field of information systems (IS), as so called “basket of 8”.

The searching keywords were used for preliminary review were written as “HUD/automotive systems/Technology + a certain terms or keywords”, for instance, “AR head-up display; LED dashboard in vehicles HUD in vehicles; benefits shortages of HUD system; functions-button on the steering wheel; safety + HUD system in vehicles; user's perspective of HUD system” etc. The inclusion and exclusion criteria were then set to assure relevant and of good quality research papers. The selected studies included those essential factors: in the English language, published between 2000-2022, and available as full text. After searching with this kind of set filters, the appropriate, relevant, and reference-worthy literature was found.

Following, the outcome of the literature review is presented as the main concepts, which create the theoretical basis for the master's thesis study.

2.2 Human-Computer Interaction

Human-Computer Interaction (HCI) focuses on interactivity and usability in computer technology, and examines the interfaces between users and computers to improve the user experience (UX) (Dix et al., 2004). In the research field of informatics, HCI has expanded to cover almost all forms of information technology to examine system's interfaces for users to

access and obtain information (Hochheiser and Valdez, 2020; Zhang and Li, 2004). Research on HCI in IS/IT could optimize the channel to access information, improve the efficiency and quality of information available to users.

In the automotive sector, HCI refers to the process of exchanging information between humans and automobiles through the interaction of, for example, buttons, knobs, voice, HUD, holographic projections, and so on (Burnett, 2009; Llanos, 2013; Schmidt et al., 2010). Since the car's mechanical controls and electrical systems have gradually been transformed into the digital domain, the car has become a complex interactive system (Schmidt et al., 2010). Thus, it is common for the driver to operate the vehicle while interacting with various in-car devices and applications. However, too much in-vehicle human-computer interaction may increase driving risks (Burnett, 2009; Schmidt et al., 2010). A trial-and-error approach with products already on the market is unacceptable in the automotive industry compared to other sectors, as the cost of failure can be fatal (Burnett, 2009). Therefore, interaction design principles are suggested to be applied instead of a trial-and-error approach.

2.3 In-Vehicle Information Systems

The in-car information system consists of the vehicle information display system and the information and communication system (Stevens et al., 2002; Harvey et al., 2011; Ozsoy et al., 2015). In this case, information on the vehicle's operating conditions can be obtained by observing the dashboard display, while information from the outside world needs to be obtained by means of communication equipment linked to the outside (Stevens et al., 2002; Ozsoy et al., 2015).

Modern vehicle information display systems use on-board computers, condition monitoring components, and other electronic instruments to provide real-time monitoring of vehicle status information (Stevens et al., 2002; Harvey et al., 2011; Ozsoy et al., 2015). The displayed information includes fuel consumption, speed, engine rounds, and temperature, which are not displayed when not required but can be retrieved by the driver by pressing the relevant button (Ozsoy et al., 2015). With the development of electronics and the application of high-tech electronic technology in cars, the information display function of future cars will be greatly expanded. More and more intelligent systems and components are being added to onboard systems. As a result, vehicle information display systems' monitoring capability and computing power have significantly increased (Stevens et al., 2002; Harvey et al., 2011; Ozsoy et al., 2015). Nowadays, vehicle information display systems can capture broader and richer information about various details of the vehicle to indicate the safety system operation state, such as tire pressure, brake device, airbag, electronic seat belt, etc. It can accurately and comprehensively determine the operating status of these safety devices, make fault diagnoses, and give fault indications to alert the driver or inform the maintenance staff of the fault (Stevens et al., 2002; Harvey et al., 2011; Ozsoy et al., 2015).

The automotive information and communication system refers to a device for communication between the car and the outside world, mainly including in-vehicle multimedia system, driver information system, language system, intelligent transportation system (ITS), global positioning system (GPS), computer network communication system, in-vehicle short-distance wireless communication system, condition monitoring and fault diagnosis

system (Stevens et al., 2002). The main components of an automotive information and communication system are rovers, mobile phones, e-mail, and international Internet terminals, video or electronic game center consoles (Harvey et al., 2011). A complete automotive information and communication system has the function of assisting the driver, the car, and the outside world as well as coordinating the electronic control of the entire vehicle components, enabling the driver to receive a large amount of information, thus improving the comfort of the driver's operation and the safety of the vehicle's movement (Harvey et al., 2011; Ozsoy et al., 2015; Stevens et al., 2002).

2.4 Automotive Head-Up Display Systems

The automotive head-up display (HUD) system is widely defined as a driving data projector with a transparent display, which presents the information at the centre field of the driver's front vision (Liu and Wen, 2004; Park et al., 2013; Wang et al., 2018; Betancur et al., 2018; Beck, 2019; Park and Park, 2019). While using the automotive HUD system to access information, drivers can keep their eyes on their usual view area and see outside directly through the HUD display without refocusing or moving (Bergman, 2011; Park et al., 2013; Ishizaki et al., 2014; Smith et al., 2015; Pauzie, 2015). Figure 2 shows a realistic driving scenario of a modern car equipped with an automotive HUD system. As the green circle marked, the HUD provides various driving information, such as current speed and road speed limit, navigation map, and vehicle status.



Figure 2. Automotive HUD system in action (author's photograph)

While the driver needs to carry out secondary tasks, such as navigating and phone call, the HUD system offers the required information for driver as a practicality and convenient user-interface (Park et al., 2013; Ishizaki et al., 2014; Pauzie, 2015; Wang et al., 2018; Betancur et al., 2018; Beck, 2019; Park and Park, 2019). As Figure 3 shows, the traditional head-down display (HDD) usually cases a visual-down angle of 17 degrees. The image is presented at a distance of 0.5-0.8 meters from the driver. Compared to the HDD, the automotive HUD system only requires a visual angle of approximately 4 degrees, the distance is more than 2 meters from the driver which is closer to the road environment (Liu and Wen,

2004; Ablaßmeier et al., 2007). The application of automotive HUD system, thus, reduces the time that the driver's eye off the road, and ensures the performance and safety of the primary driving tasks.

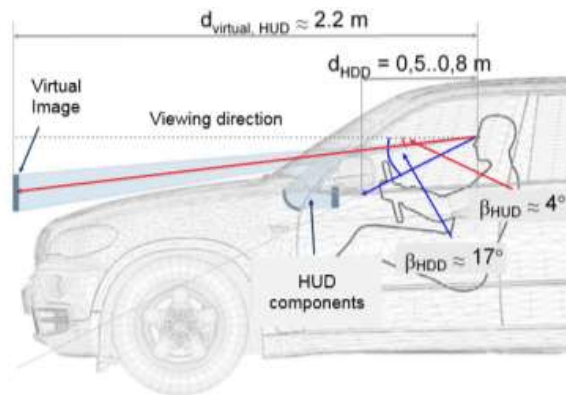


Figure3. Automotive HUD system vs Traditional HDD (adapted from Götze and Bengler, 2015)

A HUD system usually consists of three main components: 1. graphics generation computer, 2. projector unit, and 3. Combiner (Wood, Howells and Spitzer, 2001; Park and Jung, 2014). The graphic generation computer controls the image source, converting vehicle status, driving status, road information, and user-requested data into visual data and transmitting them to the projection unit for presentation (Park and Jung, 2014). The projector unit is an image information output device. Modern commercial vehicles usually install the projection unit behind the driver-side dashboard, as indicated by the yellow markings in Figure 4. The graphic display of this HUD's projection unit is shown as the green mark in this figure.

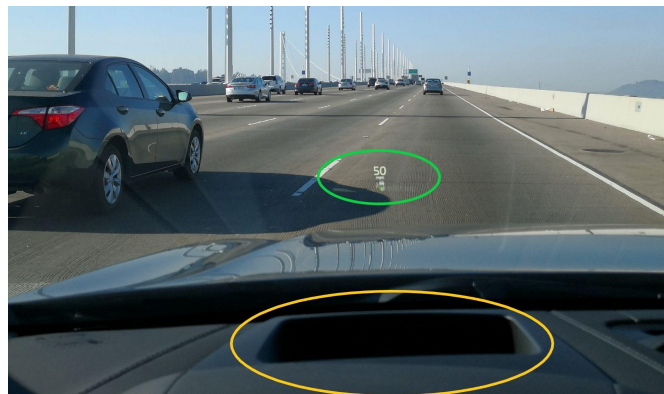


Figure 4. Projection unit of automotive HUD system and its display (author's photograph)

The combiner is a transparent presentation plate where the visual information is placed (Wood, Howells and Spitzer, 2001; Park and Jung, 2014). In the modern commercial vehicles, the combiner is embedded and integrated into the front windscreen as a lamination which allows the HUD system to present information directly on the windshield (Wood, Howells and Spitzer, 2001).

2.4.1 Technical Base: Augmented Reality

Today's advanced HUD systems are based on Augmented Reality technology (George et al., 2012; Gabbard, Fitch, and Kim, 2014; Lee et al., 2015; Betancur et al., 2018; Wang et al., 2018). Augmented Reality technology is usually considered to be a live view of the real world, onto which virtual information is superimposed, and can be interacted with (Milgram and Kishino, 1994; Kipper and Rampolla, 2012). With technological breakthroughs in optics, AR, and image recognition, AR-HUD is presented in front of the driver in a most natural way of image processing which allows these images to represent the real world and provide more useful information, such as navigation, and threat recognition (Gabbard, Fitch and Kim, 2014). Displaying information directly on the real road is an important feature of the AR-HUD. As a result, the AR-HUD system brings visual information intuitively while driving (Lee et al., 2015; Jose, Lee and Billingham, 2016). It optimizes driving behavior when AR-HUD tends to present cues that are relevant to the driving task. However, it reduces the allocation of visual attention and leads to distractions in decision-making before the driver is required to perform the task once the AR-HUD is not specific to driving-related (Gabbard, Fitch & Kim, 2014; Smith et al., 2015; Chen et al., 2016; Beck and Park, 2018; Wang et al., 2018). An effective AR-HUD system should thus only provide salient or prioritized information directly relevant to performing the driving task.

The AR-HUD is directly related to the driver's reasoning and conceptualization of strategic tasks, and the future AR-HUD interaction design and user experience will largely influence driving safety (George et al., 2012; Lee et al., 2015; Wang et al., 2018). The AR-HUD still leaves a vast space for optimization in terms of information interaction and technology. (Gabbard, Fitch & Kim, 2014; Beck and Park, 2018; Beck et al., 2019). The design aesthetics of the AR-HUD are not necessarily conducive to driving safety. Therefore, before AR-HUDs can be mass-produced, Human-Computer Interaction designers still need to work hard and always remember to design for 'driving safety'.

2.4.2 Interaction

Many information systems use the HUD systems as output interfaces in the automotive environment (Charissis and Papanastasiou, 2010; George et al., 2012; Park and Kim, 2013; Guo et al., 2014). Same as the traditional graphical user interface, HUD transmits visual information to the user through visual displays and feedback to conduct information exchanges. For inputs, on the other hand, the driver can interact with the HUD system via a touch screen which is integrated into the centre console, to select various functions of the HUD system, control outputting contents and settings (Doshi, Cheng & Trivedi, 2008; Charissis and Papanastasiou, 2010; Yang et al., 2016). In addition, the multifunctional buttons on the steering wheel allow drivers to conduct quick and easy control of the HUD, such as playing and pausing music, answering or rejecting calls (Pauzie, 2015; Choromański, Grabarek, and Kozłowski, 2019; Prabhakar et al., 2020). Advanced smart cars even feature a voice recognition system that allows the user to give voice commands to the HUD system (Charissis and Papanastasiou, 2010; George et al., 2012; Gabbard, Fitch, and Kim, 2014; Pauzie, 2015). Furthermore, researchers have explored haptic feedback, gesture recognition, and eye-tracking to facilitate human-computer interaction to enhance a new type of intelligent

interactive HUD system. This smart HUD system is controlled by tracking the operator's eye gaze or finger movements to optimize the interaction (George et al., 2012; Gabbard, Fitch, and Kim, 2014; Pauzie, 2015; Prabhakar et al., 2020).

As aforementioned, HUD is practically an interactive in-car system, and users interact with it through its interfaces (Doshi, Cheng, and Trivedi, 2008; Schmidt et al., 2010; Beck et al., 2019; Prabhakar et al., 2020). Those interfaces can either be the graphic interface of the HUD system itself as an output or interact through the interfaces of other in-car systems as the medium (Prabhakar et al., 2020). In the current stage of automotive HUD systems, the widely applied interaction interfaces are through the multifunction steering wheel and the central control system, which operate the input to the HUD system. For output, in turn, by its own graphical interface to transmit the information to the drivers (Prabhakar et al., 2020). When studying how drivers use these interfaces and thus interact with the HUD system, an important indicator is their properties, i.e., considering their usability, user experience, security, and adaptability (Schmidt et al., 2010). The properties of these interfaces directly influence the quality of the interaction greatly. Without violating human usage habits and behaviors, the excellent properties of these interaction interfaces represent a successful design of the interactive system (Schmidt et al., 2010). Interface properties are therefore a key focus when examining the interaction between the driver and the vehicle's HUD system.

2.4.3 Information

Providing a clear definition about information is not easy because there is not a clear distinction between information and the related terms of data and knowledge (Chatzipanagiotou, 2017). Data are signs which are used to represent information; information is data that require an interpretation process in order to be turned into information, and knowledge is contextual information which after its evaluation guides people's actions. Thus, information in this master's thesis is defined as the object of transmission and processing of audio, messages, and communication systems. According to Chatzipanagiotou (2017), information requires swift attention, analysis, and action within a limited time.

The current mainstream AR-HUD products contain many information functions, including but not limited to: the current speed of the car, ACC adaptive cruise assist, distance warning, lane change alert, ambient pedestrian warning, lane departure warning, and forward warning, which are not identical in the way they present navigation information (Harvey et al., 2011; Ishizaki et al., 2014; Pauzie, 2015; Smith et al., 2015). AR-HUDs can also contain complex visual information: luminance elements, graphic elements, colour elements, layout elements, crowding levels of information, etc. The visual elements of the interface for these products use highly saturated colours of red, yellow, blue and green and are in line with colour semantics, e.g. red for immediate threat symbols, yellow for slight warning symbols, and blue and green for symbols regarding driving information (Stevens et al., 2002; Charissis and Papanastasiou, 2010; Park and Kim, 2013). It can be explained as the information presented in the AR-HUD being presented in a simple and straightforward manner, with information displayed in a prioritised manner. No redundant information is thus displayed and only the most valuable information to the user shows up on the interface, which notably improves the efficiency of the driver's information access and also achieves the goal of not disrupting the integration of the environment and the interface, making it more suitable for the AR-HUD's

interface design(Stevens et al., 2002; Charissis and Papanastasiou, 2010; Park and Kim, 2013).

The automotive HUD system, on the other hand, transmits the necessary driving information visually to the driver and helps them to make better driving decisions(Park and Kim, 2013). In turn, the driver can also operate and control the HUD system display to select the required information according to the needs(Prabhakar et al., 2020). In this case, the information thus becomes the bridge between the driver and the in-car HUD system(Beck. and Park, 2018.).

2.4.4 Safety

Safety is a constant topic in the automotive sector. One second behind the wheels can mean life or death. The interaction design of the automotive HUD system is therefore based on the principle of "safety first". In particular, the interaction concept of the HUD system is to keep your eyes on the road and your hands on the steering wheel(Beck, D et al., 2019; Park and Park, 2019; Prabhakar et al., 2020). The driver's gaze can always be kept on the road ahead while using the HUD system to obtain various visual information. When a quick control or operation of the in-car HUD system is required, the driver drivers interact with the HUD system through the physical contact of the multi-function buttons on the steering wheel, their hands remain in control of driving at all times without moving hands away(Choromański, Grabarek, and Kozłowski, 2019; Prabhakar et al., 2020). For instance, when the drivers have to carry out secondary driving tasks such as playing music, checking the navigation, or answering the phone, they only need to look ahead normally and click the buttons on the steering wheel. This form of interaction with the HUD system allows these secondary tasks to be performed efficiently without any interrupting or impacting the primary driving task(Beck et al., 2019). This form of interaction ensures maximum driving safety.

However, some safety gaps are associated with the automotive HUD system. Firstly, due to the limited performance of the multifunction steering wheel, it can only provide quick operations to the HUD system(Charissis and Papanastasiou, 2010; Guo et al., 2014; Choromański, Grabarek, and Kozłowski, 2019). Many of the control interfaces for the HUD system are placed in the car's centre console, which means that the input interaction with the HUD system cannot fulfill the purpose of keeping the hands on the steering wheel(Guo et al., 2014). In addition, the visual information outputs of the automotive HUD system must be clean, simple, and intuitive to minimize unnecessary distractions to the driver(Park and Kim, 2013; Smith et al., 2015). Once the visual information presented by the HUD system is too complex or messy, it may cause information stacking and information overload, which in turn could interrupt the driver's concentration and increase the threats to driving(Park and Jung, 2014; Smith et al., 2015; Park and Park, 2019). Therefore, the automotive HUD system cannot present unlimited information, and its graphical interface must be clean, tidy, and simple, in order to meet the safety principles(Park and Kim, 2013; Smith et al., 2015).

2.5 Theoretical framework: User-Centered Design

The concept of User-Centered Design is simple: take the user into consideration at every step in the development of a product (Gulliksen et al., 2003; Mörtberget et al., 2010). More specifically, it's trying to optimize the user interface based on how the user performs the task, wants the task, and needs the task, rather than forcing them to change their habits to fit the designer's thoughts (Gulliksen et al., 2003). User-Centered Design can be described as a multi-stage problem-solving process that requires designers not only to analyze and predict how users are likely to use the product but also to test their assumptions in real-world usage situations with real users (Mörtberg et al., 2010). Figure 5 below presents a simple process of User-Centered Design, created by the author. As it shows, the colored squares from left to right represent the individual steps. First is to understand the context of use, then define the user's requirements. After that, propose a design solution. Finally, evaluate against requirements. The evaluation results are fed back into the previous steps for optimization and improvement. The whole process can be repeated. As shown by the circle in this figure, the master's thesis is placed in the phase of evaluating the users' requirements. In other words, based on the master's thesis research purpose, the study explores the interaction between the user (car driver) and the interactive system (HUD). That is, the research evaluates the users' requirements in regard to their interaction with the interactive system, and whether this interaction meets the interface properties of user experience, usability, and security. By getting the users' suggestions, the aim is to improve the interaction with the interactive system. That is, how the participants value diverse user interface properties and how they manage potential conflicts between them.

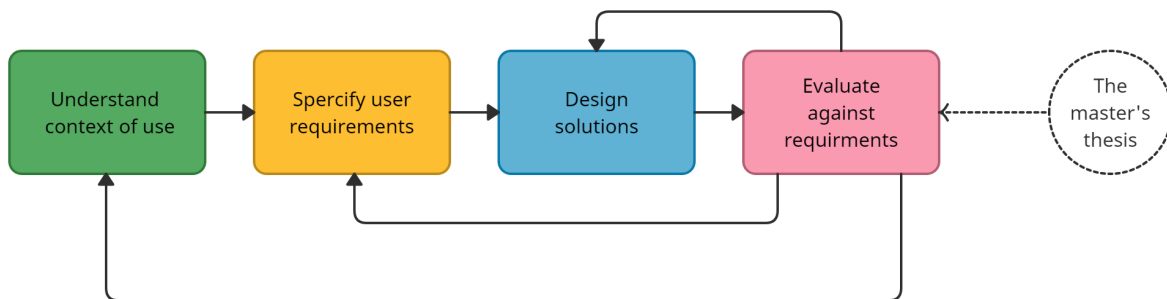


Figure 5. Projection unit of automotive HUD system and its display

The literature review, which includes key concepts from the information systems research field such as augmented reality, head-up display technology and related systems, and information and safety as the main purposes of such a system create the theoretical basis for the master's thesis research study. The aforementioned literature along with the theory of User-Centered Design form the theoretical framework of the master's thesis, which is used to interpret and discuss my research findings. The research findings was then be discussed in the discussion chapter in relation to the research aim, research questions, and will be discussed and explained with the help of the theoretical framework.

3. Methodology

In Chapter 3, the research design of this master's thesis is presented. After discussing the commonly used paradigms in IS research, the one most suitable for this research has been selected and justified. Subsequently, the followed methodology, and the data collection and data analysis methods of this research are presented and motivated. Finally, the validity and reliability in addition to ethical considerations are discussed.

3.1 Philosophical Tradition (Paradigm)

Paradigms are a set of fundamental assumptions that derived from the world about the nature of knowledge and existence. As the research strategies and criteria for rigour, paradigms are recognized and shared by the scientific community to guide the study to an appropriate direction(Myers, 1997). Choosing a paradigm appropriate to the research setting and expressing knowledge is fundamental to guarantee a solid research design(Lichtman, 2013).The main paradigms in the informatics/information systems research field are: critical, positivist, and interpretive (Myers and Avison, 2002).

The critical paradigm considers that social reality is produced and reproduced by people and is historically constituted. Even though people can have an impact on their social and economic circumstances, their ability to change it is constrained by cultural, social and political domination. Critical research is interested in the conflicts, oppositions and contradictions in contemporary society, and tends to be emancipatory(Myers and Avison, 2002). This study focuses on the phenomenon during driving, explains the benefits and challenges of using the HUD system, and then provides users' perspectives as the recommendations. It yet does not possess any evaluation or judgment of this technology. This research thus doesn't fit the critical paradigm.

The positivist paradigm is about considering reality as objectively given and described in models by properties that are measurable, independent of the researcher and his instruments. In addition, research within the positivist paradigm should be quantifiable so that statistical analysis can be performed (Myers, 1997).

The interpretive paradigm is about understanding phenomena through the meanings people give to them(Myers and Avison, 2002). This paradigm declares that reality can only be accessed through social constructions such as shared meanings and language (Myers and Avison, 2002). According to Orlikowski and Baroudi (1991), interpretive researchers rely on the participants' views of the situation being studied and recognize the impact depending on the participants' own experiences and background on the research. This study is based on observations of users' driving with HUD systems. Interviews are conducted to obtain their subjective understanding and analyze the HUD system in vehicles from the user's perspective. Therefore, the interpretive paradigm is suitable for this master's thesis research.

3.2 Research Approach and Methodology

Each of the research paradigms not only constructs a foundational framework of the worldview about the nature of knowledge but also provides an appropriate research approach. There are three approaches in research: the quantitative, the qualitative, and the mixed methods approach (Creswell and Creswell, 2017). Quantitative research involves numbers, logic and an objective stance, focuses on unchanging data and detailed convergent reasoning (Creswell and Creswell, 2017). Qualitative research, in another hand, is not a single approach but a collective term for many different research methods. As none of them are regarded as quantitative research, they are classified as the same exploration category. The purpose of qualitative research is to gain a more in-depth understanding of human behavior and its rationale. In contrast to quantitative research, qualitative research focuses on a smaller but more focused sample to generate information or knowledge about a particular study case. The mixed methods approach combines elements of both quantitative and qualitative research to answer research questions. Compared to the independent quantitative or qualitative research approach, mixed-method can help researchers get a more complete picture because it combines the strengths of both approaches (Creswell and Creswell, 2017). The master's thesis focuses on investigating the reasons and practices of human decision-making, rather than just what decisions people make or when or where they make them. In particular, by understanding how drivers interact with in-vehicle HUD systems, analyzing the motivations and factors behind this mode of interaction, and incorporating driver perspectives to draw a systematic conclusion. This fits in with the ethnographic approach in quantitative research.

In order to further address the fundamental and action lines of this qualitative research, ethnography is chosen as the research methodology (Crang and Cook, 2007). Ethnography originates in anthropology. However, it has been adopted and used in the IS field as well by applying ethnographic types of shorter duration and more focused, such as focused ethnography (Crang and Cook, 2007; Chatzipanagiotou, 2021). Ethnographic research allows the researcher to observe and record the details of the research topic, the researcher thus obtains a deep understanding of the phenomenon under-study (Crang and Cook, 2007). Therefore, this research is ideal for providing information systems researchers with a comprehensive insight into the examined aspects. The master's thesis focused on exploring the perspectives and experiences of drivers, presenting their regular interactions with the automotive HUD system, identifying the benefits and challenges while driving with an in-car HUD system, and discussing the potential demands for improvements concerning the drivers' perspectives. In general terms, it's the study of specific behaviours of a specific population, which is suitable for a small but in-depth studying size. Aforementioned, this thesis applied to a focused ethnography methodology in quantitative approach.

3.3 Methods of Data collection

The chosen interpretive qualitative ethnographic study entails the following methods of data collection: passive observations and individual semi-structured face-to-face interviews. The following sub-sections firstly discussed the sample size, i.e., the drivers/participants for which the interaction behaviors and perspectives were collected as data. Subsequently, the practical methods of participants' observations and interviews were identified. Finally, the specific

procedure of the observations and interviews were given. The following sub-sections firstly discussed the sample size, i.e., the drivers/participants for which the interaction behaviors and perspectives were collected as data. Subsequently, the practical methods of participants' observations and interviews were identified. Finally, the specific procedure of the observations and interviews were given. In short, they described the criteria for who the participants providing the data were, the observation and interview methods used to obtain the data, and the actual implementation procedures.

3.3.1 Sampling Technique, Criteria and Size

The selection of the participants was based on a purposive sampling (Creswell and Creswell, 2017). That is, the participants must fulfil certain characteristics to be included in the research in order to provide the researcher with rich data. More specifically, the research participants were chosen based on the following criteria: Possess a Swedish or EU driver's license; have more than two years of driving experience; be between the ages of 24 years-old up to 35 years-old. Therefore, I concluded to 6 research participants that fulfilled the set criteria and accepted to participate in the research.

3.3.2 The method of Observation

Observation is a commonly used method of collecting data in qualitative research and, specifically, in ethnographic studies . There are several types of observation, one of which is the passive observation. In passive observation, the observer passively observes the participants without interacting with them, leaving them in their natural setting and watching their ongoing behavior (Crang and Cook, 2007).

I decided to conduct passive observations during a drive with a car equipped with an automotive HUD system. Each of the participants was driving every time the care, while I was sitting on the co-driver's seat. This choice was made because I was able to carefully watch how the participants interacted with the HUD system through its interfaces. I deliberately chose to passively observe them because of safety reasons during driving. During the observation sessions, I kept my notes in a notebook.

3.3.3 The method of Interview

Interview is an other common method of data collection used in qualitative research. During an interview, the researcher asks the participants open-ended questions of a particular topic in order to collect data (Myers and Newman, 2007). When asking or communicating with the participants, the researcher must maintain a neutral perspective to avoid interfering with the participants' own logic and opinions, and allow them to freely express their experiences, opinions, feelings.

I chose to conduct face-to face semi-structured interviews with the same participants as those I observed. Semi-structure interview means that I have prepared in advance an interview

guide (see Appendix B) with main points of discussion. The interviews were conducted with each participant after the observation session, which was conducted while driving. The interview, right after the observation was completed, allowed me to focus on certain parts that were observed and required further discussion with the participants. This discussion points along with the points I have included in the interview guide helped me to have a deep discussion with the participants about their experiences in regard to their interaction when using the automotive HUD system. Benefits and challenges of this use were discussed. In addition, through the interviews I was able to discuss with them their suggestions for improving the interaction between the car driver and the automotive HUD system. The interviews were audio recorded with the participants permission (see Appendix A, the Informed Consent Form).

Thus, the observations created a basis of understanding about the phenomenon, in my case the interaction of car driver and automotive HUD system, and helped me identify points of discussion. The interviews allowed further collection of rich data through the discussions with the research participants on the identified points complemented by points of interest that I include in my interview guide.

3.3.4 Observation and Interview Procedure

The data collection process was conducted on April 8-11, 2022. The schedule within this period suited the participants' individual timetables. In Table 1 below, the date, duration, and identifications of each observation and interview are presented. For confidentiality reasons, the participants' names are replaced with numbers.

Table 1. The arrangement with research participants

Participant	Date	Duration(observation+interview)
1	8.04.2022	30 minutes + 60 minutes
2	9.04.2022	50 minutes + 30 minutes
3	9.04.2022	50 minutes + 30 minutes
4	10.04.2022	40 minutes + 56 minutes
5	10.04.2022	30 minutes + 40 minutes
6	11.04.2022	60 minutes + 40 minutes

The used car was offered by a volunteer and the equipped HUD system was a typical AR-HUD available on the automotive market. In other words, the HUD system in that car had standard and essential functions.

As previously stated, six participants participated in this research. Interviewed each participant right after the driving observation of that participant was completed. Each on-road observation was maintained for 30 minutes to 60 minutes. I sat in the passenger seat and

observed each participant driving with the HUD system passively and quietly. A hand-note of participants' interactions was written during this period. After that, a semi-structured interview with each participant was requested to reflect on their experiences regarding their interaction with the automotive HUD system. The interviews were conducted in English, within a duration between 30 minutes to 60 minutes, and, with participants' permission, it was audio recorded. After six observation sessions performed while driving, followed by interviews with the same six participants, I reached saturation and, thus, decided to complete the research without having to include more participants. as no additional insightful perspectives were provided after the sixth interview.

3.4 Method of Data Analysis

Data analysis with sound logic is essential to gain insight into the phenomena that the collected data represent (Braun and Clarke, 2006). Thematic analysis is often used for qualitative research (Braun and Clarke, 2006). The purpose of thematic analysis is to identify the significant or exciting data themes, interpret and understand the phenomenon behind those themes, address the research question, and present a view on it. Thus, thematic analysis was chosen to analyse the collected data of this master's thesis research. According to Braun and Clark (2006), thematic analysis consists of the following steps:

Step 1. Get familiar with the data: The first step includes transcribing the recorded interviews and gathering the written observation notes. I brought together the interview transcriptions and my observation notes, and I read the collected data several times carefully.

Step 2. Generating the first codes: The second step is to identify the initial codes in the collected data. So, after reading the collected data several times, I identified words or sentences or parts of the text, which seemed meaningful and repetitive by the participants. I extracted the list of initial codes to a separate document, where I checked them. I merged some codes, I deleted some others which with a second review were not relevant to my research purpose. This process was repeated some times until I concluded to the final list of codes.

Step 3. Exploring themes: The third step involves creating categories to place the codes written down in the previous step. So, continuing I tried to put these codes included in the final list under categories. In order to do that, I grouped them to initial categories, which, after some iterations, they were finalised and expressed in the form of themes/patterns.

Step 4. Reviewing the themes: The fourth step included evaluating, filtering, and editing the themes further. So, I did and, therefore, some themes were restructured or even removed as inapplicable; and new themes also emerged from others. Finally, the shortlist of themes was eventually formed.

Step 5. Name Giving and further defining: The fifth step involves analyzing and interpreting each theme systematically, then summarize each theme with a name. So, I gave to the final list of themes specific headings and I analysed them supported with quotations from the participants and notes from my observations.

Step 6. Produce the report: The final step involves using all the themes to construct a systematic exposition as the findings of this study, which is then brought into an appropriate theoretical framework for further discussion. Therefore, I presented my research findings under themes, and at a later chapter I discussed my findings in relation to the theoretical framework of the study.

3.6 Validity and Reliability

In the research field of informatics/information systems, meticulous attention to validity and reliability is key to assessing the quality of the research (Klein and Myers, 1999). It helps to ensure that other scientists accept the findings as credible and trustworthy to apply them in the service of practice. Therefore, a comprehensive assessing of the study's validity and reliability are important. Klein and Myers published seven principles to assure validity and reliability of the research (Klein and Myers, 1999). These principles were selected and used as criteria at the researcher's discretion to evaluate and judge the validity and reliability of this master's thesis research. The following principles were selected:

The *Fundamental Principle of the Hermeneutic Circle* suggests that all human understanding of the whole is established by reference to the individual parts. Vice versa, the understanding of each individual part is established by reference to the whole (Klein and Myers, 1999). An iterative data analysis followed a combination of considering every single part of the data and its totality. Specifically, the analysis of this master's thesis examined any part of a single observation and the complete situation of this observation; each observation of an individual participant together with the overall performance of all participants. Similarly for the data collected from the semi-structured interviews. The analysis considered any part of a single answer and the complete response to the question; the relationship of all the responses from a single participant and the growing understanding gained from the review of all interviews' material. Thus, both a detailed understanding of a single data fragment and the overall complexity was obtained.

The *Principle of Contextualization* highlights the importance of critical reflection on the social and historical background of the research setting (Klein and Myers, 1999). Since this setting was regarded to be similar to the researcher's perception and knowledge area, the analysis may possess a particular personal subjectivity. Therefore, this master's thesis ensured that the interpretations were drawn from the evidence only, the additional meaning from prior personal experience was not discussed. At the same time, the study exposed the interaction through observation and interview, which is inherently contextualized. The findings of this master's thesis, thus, were critically reflected as related elements of the research setting (the car itself). However, qualitative research is essentially a matter of interpretation; yet one cannot be divorced entirely from prior own experiences and perceptions, which was taken into consideration.

In addition, to ensure that findings were drawn from the data only, the *Principles of Dialogical Reasoning* (Klein and Myers, 1999) were used throughout the data collection and the data analysis process of this master's thesis study. Any preconceptions arising from the literature review, or the design of the research questions were avoided. For this purpose, a data-driven approach was used.

Regarding data analysis, both the observation notes and audio recordings of the interviews were provided to guarantee accuracy and validity. The notes described the participants' behaviors and interactions while driving the car equipped with the HUD system. The texts were written as straightforwardly and accurately as possible, without unnecessary adjectives. Subsequently, the observed situations on the notes were then used as the foundation for certain interview questions, allowing the participants to elaborate their own explanations. The

audio recordings were generated for each participant in order to listen repeatedly for exploring potential factors. Furthermore, the approach to data analysis was strictly following a systematic strategy under the guidance of thematic analysis and, later discussed with the literature review.

The validity of the ultimate research findings can be assured by a number of actions on the part of the researcher. However, the validity of the research findings for other similar contexts, industries, technologies and any other specific cases can only be assessed by interested potential readers.

3.7 Ethical Considerations

According to both the *Ethical Considerations in Qualitative Research*(Munhall, P.L., 1988), *Doing Ethnographies*(Crang and Cook, 2007), and *Ethical Considerations in Research Studies*(Connelly, L.M., 2014), a number of ethical aspects were considered. All participants were informed of the research purpose, expected activities, and the conditions at the invitation of participation through a written informed consent form, which can be found in Appendix A. The participants participated voluntarily by signing the informed consent form to formally consent to participate in driving observations and interviews, to collect relevant data for use in the research of this master's thesis. Their participations in the study was confidential, and the names of the participants were fabricated in the final written master's thesis.

In addition, the model and brand of the vehicle used in this research was confidential as well. The technology and applications discussed did not involve any commercial or organisational confidentiality. Avoidance of conflict or aggravation.

4. Empirical Findings

Chapter 4 commences with an overview of the findings that briefly summarizes the themes generated from the data analysis and notes some general aspects of the analysis setting and consideration as interpreted from the participants' reflections. Subsequently each theme is presented in detail together with the evidence that supports those findings, and a short summary of findings is given in the end.

4.1 Overview of Findings

The analysis of the collected data was based on the participants, i.e. the car drivers' experiences in regard to interaction when using an automotive HUD system, and their suggestions for improving their interaction with the respective system. The themes that emerged from the analysis of the collected data are the following:

Theme 1: Forms of interaction

Theme 2: Setup and adaptation

Theme 3: Reducing the usage of dashboard and centre control display

Theme 4: Less likely to affect gaze and driving habits

Theme 5: Graphical interface

Theme 6: Suggestions to improve the interaction

The meanings of these themes are presented and further elaborated in separate sections below with the support of transcribed quotations from the participants and the observation notes. The themes are presented in a narrative-flowing format to better understand the findings.

According to the sampling criteria described previously, the volunteered participants all met the minimum requirements, which means the participants had safe driving skills and essential vehicle knowledge for this research. However, some of them had a remarkable driving experience and automobile backgrounds. Due to the varying levels of knowledge and expertise relative to cars, certain of their perspectives and driving behaviors under the in-car HUD system showed the gap. Therefore, the observed individual driving behavior and interaction performance were discussed independently, without any unnecessary cross-comparisons between participants.

4.2 Theme 1: Forms of interaction

Interaction with the HUD system is an aspect of notable importance, as it at times needs to be guided in the right direction or otherwise manipulated by the driver. It was observed that the participants interacted with the HUD system by using their fingers to operate the buttons on the multifunction steering wheel and the touch screen on the centre console to set, adjust and select the functions of the HUD graphical interface. This then prompted the system to output

visual information from the HUD for the driver to consider in order to make driving judgments and decisions.

All participants set up their navigation via the touchscreen on the centre console before the drive started. Some of them connected the mobile phone with the centre console to get additional HUD information displays: information about the songs played and caller ID. However, during the subsequent driving observation, participants were more likely to use buttons for input controlling and received HUD's visual information. They hardly ever interacted through the touch screen again. *"Generally, you won't use this touch screen after setup, the rest of the interaction is just through your eyes and the buttons on the steering wheel"*, said Participant 6. To this perspective, participant 5 added the following :*"Unless you really need to make big changes to the HUD's functionality or display during your driving, no one will touch it if nothing else happens."*

Participants also considered that the interaction with the HUD system was not mandatory. As participant 1 and 2 said: *"I only look at it when I need to, if I don't, I just ignore it"* *"I know it was trying to talk to me, but I just didn't listen."* The same situation was observed during the driving of participant 3. It was clear that Participant 3 exceeded the speed limits, and the HUD began to warn him/her of this danger. However, Participant 3 did not slow down and maintained his/her high speed. This means that Participant 3 ignored the warning. Participant 3 said: *"Yes I know, but that's how I wanted to drive at the time, so I just didn't look at it"*. Participant 3 easily rejected the information from the HUD system.

4.3 Theme 2: Setup and adaptation

Depending on the driver, a complex system such as the HUD can be considered an inconvenience if setting it up takes effort and adapting to its characteristics takes time. For instance, Participants 1 and 4 found that it was especially hard to set up and considered the adaptation aspect to be rather time consuming. Participant 1 notes that it was annoying in the beginning but also that they were able to see the benefits it offers once they got used to it. Through my observed I also observed that participants experienced negatively the large number of buttons and they felt that they needed to try the HUD system out to figure out how it worked before actually using it while driving. Likewise, it was observed that the height of the projection screen was not favourable, and they could not find out how to properly adjust it. Participant 4 commented that it was also difficult to adjust the height since the feeling was that it was too small and too difficult to see, and even gave up on trying to figure out the system since it was so difficult to understand. Participants 5 and 6 were observed to spend some time with the centre console to adjust the system to their liking, but they were not unsatisfied with its existence overall. On the other hand, Participants 2 and 3 were instantly pleased with what the system had to offer; the size, positioning, brightness, etc. all worked well for them and they did not feel that they needed to make adjustments to the HUD system.

Overall, it was observed that the participants had to spend some time figuring out the functions of the HUD and the controls of its graphic interface ahead of their driving session. Even the ones who did not find it difficult to adapt still tinkered a bit with what was offered. Through interaction with both the centre console and multifunctional steering wheel, participants enabled navigation on the HUD system and connected their phones to the car by

bluetooth in preparation for visual information regarding music playing and in-car phone calling. Some participants even tried to adjust the graphic interface of the HUD system for a display that suited them better. This process took around a few minutes. During this period of set up, participants generated different perspectives, both positive and negative.

4.4 Theme 3: Usage of dashboard and centre control display

I observed that the HUD system minimises the need for drivers to look down and check the dashboard or centre console display for information, something which was later confirmed through the interview discussion with the participants. For example, during the driving observation, Participant 1's gaze barely ever moved down or to the centre. This indicates that Participant 1 did not check either the centre console display or the dashboard. The participants explained in the interview afterward: *"Yes, basically I didn't look at the dashboard at all, because the HUD was already telling me how fast I was driving"*, Participant 1 said and continued *"After I had set up the navigation and connected my phone, the centre console display wasn't needed any more"*. Participant 2 was observed to have reduced the frequency of dashboard checking by a great deal. *"I don't even need the dashboard when I have this system."* said Participant 2, who hardly looked down at the dashboard while driving. *"It (HUD system) gives me the most important information (speed monitor) already, so I don't need to see the same thing on the dashboard again"* and continued *"For me, the centre console display is just for navigation maps and phone call information, both of which are now available to me from the HUD"*.

Participant 6 made rapid vertical visual movements occasionally, which means he/she were still checking the dashboard briefly when he/she was driving with the HUD system. When in the interview s/he was asked about the reason for looking down at the dashboard, Participant 6 replied: *"I'm getting used to it, and I also want to see if it has the same speed monitoring as the dashboard."* Although both Participants 3 and 5 also checked the dashboard more often than others, they still believe they looked less dashboard than usual. *"It's (flexible gaze and checking dashboard) just my muscle memory."* said Participant 3, *"but I used definitely less dashboard"*. Participant 5 said: *"It (HUD system) certainly helps quite a lot, I get to know the speed and where I should go (by HUD's navigation), but it's not a must, as I still got some old habits of looking at speedometer and stuff (on dashboard)"* and added *"Yeah, of course I looked the dashboard less than usual."* Participant 4, experienced the use of HUD system differently. *"It's too small for me, I couldn't see anything."* Despite this, s/he still admitted to checking the dashboard less often than usual. *"Especially in the city, I hardly used the dashboard to know my speed"* Participant 4 commented that in complex road conditions such as urban areas, it was obviously becoming less reliant on the dashboard than before.

4.5 Theme 4: Gaze and driving habits

During driving, all participants were observed to show typical driving behaviour when they used the HUD system. Participant 1 did not move her/his eyes in any vertical direction, which also means that did not check the backside mirror. In response to the lack of looking in the backside mirror, s/he said: *"I didn't look in that mirror because I know these roads very well, so I didn't need to check my backside. If I had just been driving my own car without the HUD"*

system, I wouldn't have looked at that mirror either". Besides the vertical gaze, Participant 1, however, maintained a flexible horizontal field of view movement during the driving, which suggested that their vision was not actually locked to the front only. *"Yes, I always look at the left and right side mirrors when I drive. I have to know what is happening on these two sides"* said Participant 1. Participant 6 made both rapid vertical visual movements occasionally, which means they were checking both the backside mirror and side mirror briefly when s/he were driving with the HUD system. *"I'm getting used to glancing at all the mirrors every time I drive. I wish to know what's going on around me."* Participant 6 maintained the personal driving habits while using the HUD system.

Participant 3 and Participant 5 were extremely flexible with their gazes during their driving, remaining wandering between different angles, and hardly fixated on one visual point for too long. *"No, I wasn't affected, just driving normally"* said Participant 3. Participant 5 said: *"it has the same focus length as the road so I don't really need to focus on it, scanning over it a few times per minute is good enough I would say"*. On the other hand, Participant 6 also showed a flexible driving eyesight. However, Participant 6 had a different response in the interview: *"It caught a bit of my focus, But it was more about psychological, since you will feel there is always something. But it can be Ignored easily. Especially once you get used to it, then you can totally ignore it. Just like you look at your nose."*

4.6 Theme 5: Graphical interface

The different experiences, backgrounds and preferences of the participants led to different perceptions of the use of the HUD. Participants 1, 2, 3, 5 were observed to have enjoyed the graphical interface very much. Participant 1 loved the displays of navigation and music information. He/she expressed that the design of graphic interface was pleasant: *"It's simple and clean, nice colour and shape to know the meanings easily"* and added *"the music display was so good! you can just see if you like the next song instead of taking time to hear and recognize it first"*. Participant 2, who also voiced the love to the graphical interface, considered that the truly valuable information was conveyed through this system: *"Very nice and concise, all the important information was given to me and nothing else was superfluous"* and continued *"Especially the visualisation of phone call information, it's easy to read the cue of calling, which I preferred than a loud ringtone."*

Participant 6 thought it was overall good but still could be improved. He/she kept the car at a particular distance from the vehicle in front and played with the distance alert. *"It was very cool in the way it faded in and out. This kind of showing fits a lot to the reality of the distance between cars."* When talking about shortcomings, participant 6 suggested to improve the display of the navigation map. *"GPS is too simple. The shape of the road shown is a bit wrong and lack of advance ahead, these things affected me in understanding the map sometimes and didn't know where to make the turn"*. Participant 6 also said that he/she had been exposed to strong sunlight while driving, which caused difficulty in seeing the HUD display.

Participant 4, however, was disappointed with the graphic interface of the HUD. It was observed that the small size of the graphics made it difficult for Participant 4 to see the information. After several unsuccessful adjustments, Participant 4 wanted to give up trying. In

addition, Participant 4 said that the graphics became blurred when there was too much light outside. Even the colors were mixed with those on the road.

4.7 Theme 6: Suggestions to improve the interaction

The participants had several suggestions in regard to improving their interaction with the automotive HUD,. Participant 1 wanted to enhance the long-distance driving experience by displaying more information about the geography and surrounding facilities on the HUD. *"If too much information is bothering you, you can just turn it off."* According to Participant 1, it can be inferred that what participant 1 wanted was a quick interface to control the HUD system and its functions on and off.

Participant 3 wanted to enhance driving safety by porting the alerts that detect obstacles in blind spots to the HUD system. *"Then you don't need to move your head to check the blind spots"* said Participant 3. S/he added that the reason for this additional function was to keep the eyes on the road ahead as much as possible. Participant 4 suggested the in-car HUD system to identify and mark the potential threats on the road in order to enhance driving safety as well. Participant 6 suggested the HUD system to support the usage of third-party applications to display a more accurate navigation map on the HUD system and improve the driver's experience. Participant 6 said: *"It only displays navigation when I use the in-car navigation system on the central control, not if I use google map"*. Usually, in-car navigation systems are pre-stored with map data rather than networked in real-time. Participant 5 wanted to add a web-based function to the HUD system, transforming the HUD display into a web-enabled electronic screen. S/He believed that using the HUD system to surf the internet can help to kill time when parking and waiting for people.

4.8 Summary of Findings

Six key empirical findings emerged from the thematic analysis of the collected data. The first theme described the form of interaction between the driver and the automotive HUD system. By using the interface on other in-car systems as a medium, to control the output of the HUD system, and then visual interaction through the eyes to access this information displayed on the HUD. Secondly, the findings showed that most drivers need a certain time to set up the HUD system before driving and even adapt how to use it. This setup and adaptation were time-consuming and unfriendly to users. Furthermore, findings showed that driving a car equipped with an automotive HUD system reduced the frequency of checking the dashboard, but did not affect participants' driving habits. In other words, the use of the automotive HUD system didn't impact participants' normal driving behaviors. Meanwhile, it significantly reduced the times of looking at the dashboard when driving with it. Since automotive HUD systems primarily output visual graphic information, findings showed the participants' experiences and perspectives on using the graphical interface of this standard in-car HUD system. However, as the graphic interface received mixed reviews, this theme listed the feedback from the participants only. Ultimately, findings showed the expectations and perspectives of participants on improving the interaction of the automotive HUD system. All these empirical findings are discussed in the next chapter with the literature review, including the user-centred design approach.

5. Discussion

The first Research Question of this master's thesis read: How do car drivers experience the use of the automotive HUD system in regard to their interaction with the respective system? Themes1 to 5 of the findings answer RQ1 as visualized in Table 2 below. In addition, a systematic discussion based on the theoretical framework of user-centred design was conducted and user suggestions for improvement were listed.

5.1 Answer to Research Question 1

The first Research Question of this thesis read: *How do car drivers experience the use of the automotive HUD system in regard to their interaction with the respective system?* This question can be further subdivided into two sub-questions. Viz. *Sub-Question.1: describe the benefits of using a HUD system?* and *Sub-Question.2: describe the challenges of using a HUD system?* Themes1 to 5 of the findings answer RQ1 as visualized in Table 2 below.

Table 2. Themes answering the 1st Research Question

Research Question.1	Themes that Answer Research Question.1
How do car drivers experience the use of the automotive HUD system in regard to their interaction with the respective system? <i>Sub-Question.1: describe the benefits of using a HUD system?</i> <i>Sub-Question.2: describe the challenges of using a HUD system?</i>	Theme 1
	Theme 2
	Theme 3
	Theme 4
	Theme 5

Same as described in (Prabhakar et al., 2020) study, the interaction between drivers and automotive HUD system is primarily in the form of visual looking and finger operation. Fingers to operate the buttons and the touch screen to use the HUD system. This then prompted the system to output visual information from the HUD for the driver. Especially during the driving process, the driver's eyes can focus on perceiving the road conditions from different angles while receiving information from the HUD system. At the same time, the use of buttons on the steering wheel ensures that the driver can interact with the information provided by the HUD system while keeping their hands on the wheel. As (Park and Park, 2019) said, "Keep eyes on the road and hand on the wheel" resulting in a convenient, practical, and safe form of interaction. However, due to the limited performance of the multifunction steering wheel's interactive interface (Choromański, Grabarek, and Kozłowski, 2019), the user can only use these buttons to access simple information in the HUD, such as the phone book and music playlist. Rather complex controls and uses such as adjustment of the GUI, function switches, etc., must be operated on the touch screen of the central control. This leads that once the driver wants to make an adjustment to the function or graphical interface display of the HUD system while driving, they have to move their eyes off the road

to look at the center and remove their hands from the steering wheel. Some participants had already expressed their dissatisfaction with those types of interactions. In addition, it was observed that participants with extensive driving experience, or those who had already used the automotive HUD system before, could adapt quickly to interact with the system. However, it can be a time-consuming process for beginner or novice driver who is fresh with the HUD system to search back and forth between the steering wheel and the centre console in order to discover how to operate and use the HUD system.

As the display of the HUD system is transparent(Wood, Howells, and Spitzer, 2001; Park and Jung, 2014), it presents a graphic interface that does not require the driver to look at it, but rather the gaze can pass right through it. At the same time, the presentation of the graphic interface does not occupy the entire central field of the driver's view, but just a tiny part of it(Bergman, 2011; Park et al., 2013; Ishizaki et al., 2014; Pauzie, 2015; Smith et al., 2015). Due to these factors, the driver can easily ignore the presence of the HUD's graphic. In this user-driven human-computer interaction, the driver receives the visual information offered by the HUD fully according to personal intention. When not needed or desired, there can be absolutely no interaction. Evidence from these observations and interviews suggests that participants were, in fact, able to reject the visual information from the HUD by intentionally blocking the interaction between them and this system. For this reason, the automotive HUD system has shallow appeal to the driver's attention, and, therefore, the transition of information on the HUD system does not impact driving itself.

On the other hand, the automotive HUD system's application reduces the dependence on the information provided by the dashboard and the central control display. Nowadays, the information provided by the AR-HUD system is related to speed monitoring, speed limit indication, navigation maps, music/radio visual information, and caller ID(Harvey et al., 2011; Pauzie, 2015; Park et al., 2019). This available information is derived from the dashboard and the central control system, and is even a derivative extension of the native functions. In other words, the critical information provided by the dashboard and the central control system is intercepted and redesigned for display on the HUD system, which is thus regarded as a third information presentation screen that combines the main functions of the first two(Bergman, 2011; Park et al., 2013; Ishizaki et al., 2014; Smith et al., 2015; Wang et al., 2018). As the HUD system integrates important information from the other two in-vehicle displays, the driver can access this information directly through the hud, avoiding the necessity to look at the dashboard or the centre console display and thus reducing the use of these two traditional screens. Through observations and interviews with all participants, a discovery was made. The phenomenon aligns with this discussion regarding the HUD system, which is that it minimizes the need for drivers to look down and check the dashboard or centre console display for information. However, the data collected revealed that the HUD system might not present information clearly when the external brightness is too strong. As mentioned before, due to the overall difficulty in quickly adjusting the presentation of the HUD's graphical interface, it is hardly able for the driver to keep using the HUD system through the existing interaction behaviors during this period of bright light. Furthermore, by comparing each above participants' reviews of the graphical interface of this HUD system, the design of the HUD graphic interface was found to be generally appreciated by participants for its simplicity, cleanliness, and intuitiveness. Two factors were also summarised as causing negative feedback to the graphical interface: the size, height, or brightness of the graphics did not match the user's habits which were difficult or impossible to adjust, and the

misinterpretation of the images. Thus, some of the graphical interfaces that transmit information may require upgrading to fully optimize the user experience.

5.2 Answer to Research Question 2

The second Research Question of this master's thesis read: *What are the improvements that car drivers of cars equipped with HUD systems suggest for their interaction with the respective system?* Theme 6 of the findings answer RQ2 as visualized in Table 3 below.

Table 3. Theme answering the 2nd Research Question

Research Question.2	To Answer Research Question.2
What are the improvements that car drivers of cars equipped with HUD systems suggest for their interaction with the respective system?	Theme 6

Whether to enhance driving safety or increase driving funs is an advancement of using an automotive HUD system. It is apparent from the observations and interviews in the thesis that, the participating drivers expected the automotive HUD system could provide many extra features to improve their user experience. Through these additional functions, the driver could obtain richer information to have better decision-making.

An assumption proposed by the participant is that, some functions from other existing automotive systems could be ported to the HUD system to increase driving safety. For instance, on the sides of modern cars, sensors are equipped to detect obstacles from the driver's blind spots and alert the driver by illuminating the LED lights on the side mirrors(Burnett, 2009). The application of this system allows the driver to be informed of potential threats from the blind spot without turning the head excessively and checking purposely. In the thesis, this participant's improvement is that, if the alerts that detect obstacles in blind spots could be ported to the HUD system, it would even reduce the time of checking the two sides, since it's at the front and easier to be seen. The driver thus could receive the cue signal faster, which is in accord with the purpose of interacting with the automotive HUD system, to reduce the time that eyes off the road. Another suggestion from participants for improving the driver's experience is to access the information from the external third-party services through the automotive HUD system. Participants could not access navigation information on other platforms via the HUD system in the thesis. It displayed the GPS from the central control system only. If the driver can access additional external information directly through the in-car HUD system, the variety and form of information gained through interaction with the HUD will increase. This cross-platform interaction of information could thus improving the experience of driving with the HUD system.

On the other hand, as technology develops, drivers expect to have new features regarding automotive HUD systems in order to obtain richer information. For instance, the overlay of image recognition technology and augmented reality allows the HUD system to identify

potential threats on the road and mark them up for the driver (George et al., 2012; Gabbard, Fitch, and Kim, 2014; Lee et al., 2015; Betancur et al., 2018; Wang et al., 2018). In this case, the drivers can then receive information about driving safety that they may have ignored. Another example, with the Internet of Vehicles (IoV), the in-car HUD system can be transformed into a screen for surfing the internet to have a large amount of real-time information on road conditions, and even extra fun beyond driving (Yang et al., 2014).

As everyone has different knowledge backgrounds and preferences, the perceptions of automotive HUD systems' design and interaction approach are highly variable. By comparing the participants' perspectives in the master's thesis, a foundation can be found that the participating drivers expected to improve the interactive experience by having a variety of extra information presented on the automotive HUD system. However, over-increasing the amount and type of information without limitations may result in excessive information stacking and information overload instead, which can distract the driver's attention and cause potential threats (George et al., 2012; Gabbard, Fitch, and Kim, 2014; Lee et al., 2015; Smith et al., 2015; Wang et al., 2018; Beck et al., 2019). Therefore, the primary focus is still on security issues when considering optimizing, redesigning, or adding new graphic features to the in-car HUD system (George et al., 2012; Gabbard, Fitch & Kim, 2014; Park and Jung, 2014; Lee et al., 2015; Smith et al., 2015; Wang et al., 2018; Beck et al., 2019). At the same time, new extra functions on the automotive HUD system mean the new operations and controls. Combining Participant 1's suggestions with the interaction issues described previously, it can be deduced that a quick and straightforward interaction form may be required for the driver to operate these HUD system functions more conveniently.

5.3 Discussion within the User-Centred Design theory

By analyzing the data obtained through the observations and interviews, the way in which the participants interacted with the automotive HUD system was already well defined: drivers looked to the HUD display to receive visual information, meanwhile, they click the buttons on the steering wheel and touch the screen on the centre console to operate and control the HUD system. In this set of interaction forms, it appears evident that there are three interfaces for users to interact with in-car HUD systems: the graphical interface of the HUD system, the button interface on the multifunction steering wheel, and the touch screen interface on the centre console. As described in the theoretical framework, this thesis focused on evaluating user interactions with automotive HUD systems, exploring their user experience and perspectives of this interactive system. Therefore, these interactive interfaces were systematically assessed in accordance with the user-centred design theory.

Participants generally recognized the driving benefits generated by interacting with the HUD system through the graphical interface. Especially when checking the speed and navigation maps, they no longer moved their eyes off the road to search the driving information on the dashboard or the centre console's screen. In addition, when they were subjectively unwilling to interact with this graphical interface to obtain information, they were able to ignore it easily. Participants thus described the HUD system's graphical interface as a simple, useful, and friendly interaction. However, this does not represent the graphical interface of the HUD system is free from defects. The participants differed in their pictorial understanding of the graphical interface, particularly the navigation maps. The maps displayed by the HUD system

were very concise to avoid information overload. However, some participants expressed difficulties in understanding such an overly simplified map. They felt the map was different from the actual road. This misunderstanding could reduce the user experience. Furthermore, when driving in a bright light condition outside, the presentation of the HUD system's graphical interface can become blurred and even blinding to the driver. This situation happens unpredictably, and the driver cannot adjust the display of the graphical interface on time, which could threaten driving safety and harm the user experience.

On the other hand, participants experienced dissatisfaction when conducting the input interactions with the HUD system via the button interface and the touchscreen interface. During driving, most of the participants' input interactions were performed by using the button interface on the steering wheel. This interaction provides a powerful safety feature, as the drivers(participants) can keep their hands on driving all the time(Choromański, Grabarek and Kozłowski, 2019). Limited by the performance of the button interface, however, participants could only perform simple input operations, such as selecting the music menu on the HUD display, interacting with the phone menu, etc. When participants were performing complex operations, such as adjusting the brightness or height of the output image, or switching on/off certain functions of the HUD system, they had to interact with the touch screen interface first, and then combined with using the button interface. The participants described such input interactions as unfriendly. They had to explore back and forth between these two input interfaces before they started driving, which was a time and energy-consuming process. This process was painful, especially for those who were unfamiliar with the automotive HUD system. They spent a lot of time learning how to use the HUD system, even had to quit operating the HUD system, and pushed themselves to adapt it instead. Due to this complex input interaction, participants refused to make adjustments or select functions while driving since they had to move their hands and eyes to the central control. It is a paradox that drivers must take additional driving risks if they divert their attention to making input interactions; if they are willing to drive safely, then they must sacrifice their user experience. Besides, the interviews show that participants want to improve their interaction with the automotive HUD system by adding various other features. If this demand is to be met, the input interfaces thus need to handle more of the interaction. In this case, however, using the button and touchscreen interfaces only may not be strong enough to offer a safe, friendly, and convenient input interaction.

In summary, participants were satisfied overall with the graphical interface of the automotive HUD and described its output interaction as good usability, safety, and user-friendly. However, the input interactions for complex controlling and operating of the HUD system were considered as hard to use and a bit risky. Therefore, further optimizations can be made by improving the input interactions of the automotive HUD system.

5.4 Suggestions of users regarding the interaction

In order to get a clearer picture of drivers' (participants') expectations and needs for the HUD system, here is a list of their suggestions for improvements to the HUD system. Each bullet represents a suggestion, along with a short description.

- **Blind spots alert:** Porting the alerts that detect obstacles in blind spots to the HUD system, reduce the moving head to check the blind spots and improve the driving safety, which was in accord with the design guidelines of the automotive HUD system.
- **Threat identification:** Through image recognition technology, potential threats and risks on the road are marked via HUD system to notice drivers. However, in a complex road condition, for example during rush hour in a city centre, this function may generate too many markers, leading to information overload and possibly causing a hazard. Therefore, this function needs to be further examined.
- **Support the navigation data and information from the third party:** The current automotive HUD system only supports the display of the navigation system that accompanies the car itself. The ability to support the presentation of navigation maps provided by third-party data platforms would improve the experience when viewing the navigation on the automotive HUD system.
- **Surfing the internet:** Add internet accessibility to the in-car systems and use the HUD system as a screen for surfing. This suggestion matched the development of the Internet of Vehicles nowadays (Yang et al., 2014). If applied properly, it could significantly improve various aspects of the driving experience.
- **A wealth of information on the surrounding facilities:** Through the in-car network, information about facilities around the road, such as gas stations, tourist attractions, toilets, etc., is obtained and displayed on the HUD system. This feature could greatly expand the range of information on the HUD system available to drivers, thus helping drivers to make decisions, increasing driving enjoyment and user experience.

6. Conclusion

In this final chapter, an overall outline of the master's thesis is given, and a summary presentation of the research results is made. This is followed by a brief description of the contributions of this master's thesis. Finally, future directions for the research are proposed.

6.1 Conclusions

The master's thesis examined the interaction between user and in-vehicle information systems. The aim of the master's thesis was to explore and improve the understanding of the interaction between the car driver and the automotive Head-Up Display system. Further, the master's thesis explored the suggestions of car drivers in order to improve their interaction with the respective in-vehicle information system. From the above purpose, two research questions arose: RQ1. How do car drivers experience the use of the automotive HUD system in regard to their interaction with the respective system? and RQ2. What are the improvements that car drivers of cars equipped with HUD systems suggest for their interaction with the respective system? For this, an interpretive, qualitative focused-ethnographic research was conducted. The data was collected through passive observation and individual semi-structured face-to-face interviews with six purposively selected research participants-drivers of cars equipped with HUD system. The collected data was analysed thematically to conclude to six themes, which represent the research findings. These research findings were discussed with the help of the theoretical framework. The theoretical framework consists of main concepts from the body of research of human-computer interaction such as in-vehicle information systems, automotive Head-Up display information systems, interaction, information and safety, which along with the theory of User-Centered Design were used to explain and discuss the research findings.

The research findings showed that most users can have a good experience of using the HUD system while driving. Through the various buttons on the multifunction steering wheel and the touch screen of the centre console on the car, the user can select the display functions of the HUD system to control and interact with it. In particular, the interaction via the multifunction steering wheel allows the driver to use the HUD system while keeping hands on the steering wheel. In addition, the HUD is almost a new screen that combines important information from the dashboard and the centre console. This design reduces the need for the user to move their eyes off the road frequently in order to check some important information displayed on the centre console or dashboard, which greatly improves the amount of time that driver can focus only on the road. This approach to interaction is convenient and safe which loved and enjoyed by most of users. However, setting up the graphical interface for the HUD system can be troublesome. The user must first locate the corresponding setting interface on the vehicle's centre console, select the desired setting via the touchscreen and then adjust it via the buttons on the multifunction steering wheel. This is very difficult to accomplish while driving. Most users therefore choose to set up the HUD system before they start driving, or pull the car over to the side of road before making adjustments to the graphical interface. Especially for novice users, it may take extra time and effort to figure out where these setup interfaces are and how to use them. Furthermore, users look forward to adding various functions and features to improve their user experiences. For example, the Internet of

Vehicles(IoV). They are expecting to turn the HUD system into a display with internet access to get advanced in-car entertainment and driving pleasure. They also willing to obtaining the real-time information on the HUD's map via IoV to know more about road condition and the surrounding facilities. On other hand, automotive HUD systems are also expected to use image recognition and augmented reality technology to identify and mark the potential threats on the road. Certain existing cueing systems related to driving safety, such as blind spots alert systems, are also expected to be ported on the automotive HUD system for using, which could speed up the reception of alert information by the driver.

Finally, the research outcome is complemented by a list of suggestions to improve the interaction of car drivers with HUD systems, which were proposed by the research participants.

6.2 Contributions

This master's thesis research focuses on the interaction between the driver and the automotive HUD system, the essence of which is a research problem of human-computer interaction. Therefore, this master's thesis has a theoretical and a practical contribution. The research contributes theoretically to the existing body of knowledge of the informatics research field and, in particular, the human-computer interaction body of research. The theoretical contribution is explanatory and descriptive based on Gregor's (2006) categorization.

The research outcome has also practical contribution. As this interaction is based on an automotive Head-Up Display system consisting of augmented reality technology, the study contributes to designers, developers, ICT companies, car drivers, and other interested practitioners of similar products, who could make use of the insights and the new knowledge. Finally, as this study is established on the user's - driver in this case - perspective, it could be a valuable user experience report for designers and developers of automotive HUD systems, from which they can gain insight into user requirements and, thus, optimise it for the benefit of the driver. Based on Gregor (2006) the practical contribution of the research study is descriptive.

6.3 Suggestions for Future Research

A suggestion for future research would be to repeat the same research with a larger number of participants holding different roles/positions, such as, besides car drivers, designers, developers or other practitioners.

Another interesting suggestion would be to conduct a comparative study where the experiences of participants' interaction with the HUD system while driving would be compared with a group of users that would use a simulation of the HUD system.

Another suggestion for future research would be to explain and discuss the findings from a different theoretical approach. A final suggestion would be to conduct research that is complemented with quantitative data to examine if that would add more insight or value to the study.

To simplify the user's operation and optimize the user's interaction experience, an eye-tracking system (Prabhakar, 2020) and a voice command system (Bruce, Boehm-Davis, and Mahach, 2000) could be applied as the safety solutions. As the name implies, the eye-tracking system is constructed by an in-car intelligent sensor and the onboard computer to capture the movements and changes of the driver's eyes and pupils in real-time. By analyzing the data collected relating to the driver's gaze, information such as the driver's field and focus of view is extrapolated to automatically adjust the HUD's presentation, to offer a suitable display position, brightness, etc., back to the driver (Prabhakar, 2020). With this technology, the drivers can simply sit in the vehicle, passively letting this system sense eye. The HUD system is then adjusted automatically according to their eye changes. This system application avoids the need for the driver to operate HUD manually. For the voice command system, The driver only needs to send a voice command to the onboard computer with the needs, and the onboard computer can automatically adjust the HUD system for the driver (Bruce, Boehm-Davis, and Mahach, 2000). An interesting suggestion is thus to conduct further research on the possibility and applicability of adding those two intelligent systems to support the HUD system.

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Appendices

Appendix A: Informed Consent Form for Master's Thesis

Date: March/April 2022

Title of Research: In-Vehicle Information Systems: An Ethnographic Research on the Interaction between Driver and Automotive Head-Up Display System

Researcher: Shishengxiong Zhong, Master Programme in Information Systems, Linnaeus University
(sz222cu@student.lnu.se, +46 0765569501)

Purpose of the Research: This research aims to explore and improve the understanding of the interaction between the car driver and the automotive Head-Up Display system. Further, the master's thesis explores the suggestions of car drivers in order to improve their interaction with the automotive HUD system.

Expected from You: As a research participant, you will be observed passively by me, the researcher, while driving a car that is equipped with a Head-Up Display system. During the observation, I will be sitting in the co-driver seat and I will observe how you interact with the automotive HUD system, without me talking or interacting with you. The on-road observation will last 30 minutes. After that, we will have an interview where you will be requested to reflect on your experience regarding your interaction with the automotive HUD system. The interview will be conducted in English, it is not expected to be more than 40 minutes, and, with your permission, it will be audio recorded.

Participation and Confidentiality: In accordance with the good research practice advocated by the Swedish Research Council (Vetenskapsrådet), your participation implies the following:

- Your participation is 100% voluntary, meaning that you are allowed to withdraw your participation at any time during the master's thesis research.
- Detailed about your driving skills or personal information is not of interest for the purpose of this master's thesis study and will not be examined/discussed during either observation or interview.
- The text/audio/video generated during the observation and interview will be recorded for the purpose of transcription and textual analysis in this master's thesis study. The results of our discussions will be used for the purpose of the master's thesis study and for additional research purposes as well.
- Personal information provided by you will only be used and stored during the time of this master's thesis research. Your participation in the study will be confidential and your name will be fabricated in the final written master's thesis.

By signing this document, I agree to the terms in this consent form. Thank you for your participation, and please do not hesitate to contact me if you have any questions!

sz222cu@student.lnu.se

+46 0765569501

Date:

Date: 2022-03-16

Participant's name:

Researcher's name: Shishengxiong Zhong

Signature:

Signature:

Appendix B : Script and Questions for Semi-Structured Interviews

General Initial Question:

Please tell me about your driving experience (how many years have you been driving) and how many hours or kilometers on average are you driving per week. Do you own a car yourself? If so, what kind of car? Or, did you own a car? If so, what kind of car.

Automotive HUD Systems and Interaction

1. If I say Head-Up Display system, what comes to your mind? (are you aware of what it is?)
2. If I say automotive Head-Up Display systems, what comes to your mind?
3. Could you please tell me a couple of things about how an automotive Head-Up Display system works in cars?
4. Do you have any previous experience of driving cars equipped with Automotive Head-Up Display Systems? If yes, how long have you been using this system for driving?
5. Based on your current driving experience of driving a car equipped with a Head-Up Display system, could you describe this experience (was it difficult, pleasant, challenging etc.)? during the observation period/previous usage experience, How do you feel about the HUD system?
6. How would you describe the use of the Head-Up Display system in the car? (did you find the use easy, difficult, pleasant etc.)?
7. Based on your current driving experience of driving a car equipped with a Head-Up Display system, which features of this HUD system do you find satisfying?
8. Based on your current driving experience of driving a car equipped with a Head-Up Display system, which features of this HUD system do you find less satisfying?
9. Based on your current driving experience of driving a car equipped with a Head-Up Display system, are there any features of this HUD system that you found uncomfortable?
10. Based on your current driving experience of driving a car equipped with a Head-Up Display system, how would you describe your overall interaction with the HUD system?
11. Based on your current driving experience of driving a car equipped with a Head-Up Display system, would you suggest other services to be included in the HUD system? Why?
12. Do you have any thoughts regarding the design of the interface or information presentation of the HUD system?

Questions that could emerge from the Observation

13. During today's driving experience of driving a car equipped with a Head-Up Display system, could you please explain for what reason you focused on the HUD system?
14. During today's driving experience of driving a car equipped with a Head-Up Display system, your eyes were focused on the HUD system for X seconds. Do you think this focus time is too long or too short for you?

15. During today's driving experience of driving a car equipped with a Head-Up Display system, I observed that your hand/fingers adjusted the function selection of the HUD system by using ..., could you please tell me the reason why you needed to select other functions? What feelings do you have when you need to adjust some functions while driving?

16. During today's driving experience of driving a car equipped with a Head-Up Display system, you checked the various rearview mirrors X times, for about X seconds. Is this frequency the same as when you drive a car, which is not equipped with a HUD system?

Concluding Remarks

17. Would you like to add something else in regard to your current driving experience of driving a car equipped with a Head-Up Display system?

18. Would you like to add something else in regard to your current experience of using an automotive Head-Up Display system?

Thank you very much for your participation.



Linnæus University
Sweden

Faculty of Technology

SE-391 82 Kalmar | SE-351 95 Växjö

Phone +46 (0)772-28 80 00

teknik@lnu.se

[Lnu.se/fakulteten-for-teknik](https://lnu.se/fakulteten-for-teknik)